Weekly Practice and Troubleshooting

1. Background

1.1 WAN technologies in brief

Technology	Description	Typical use
Leased line	Dedicated point-to-point circuit with guaranteed bandwidth	Critical links between head office and data centre
MPLS / Metro Ethernet	Provider-managed private WAN with QoS	Enterprises with many branches
Broadband (DSL, cable, fibre)	Shared, low-cost Internet access	Small branches, home offices, VPN underlay
Cellular (4G/5G)	Wireless WAN via mobile network	Backup links, mobile and remote sites
Internet VPN (IPsec)	Encrypted tunnel over the public Internet	Low-cost secure branch-to-branch connectivity

1.2 Site-to-site IPsec VPN

A site-to-site VPN joins two private LANs across an untrusted network by encrypting traffic between two edge routers. It is built in two phases:

- **IKE Phase 1 (ISAKMP):** the peers authenticate each other (here with a pre-shared key) and agree on a secure management channel. Verified with show crypto isakmp sa; the healthy state is QM_IDLE.
- **IKE Phase 2 (IPsec):** the peers negotiate the IPsec security associations that protect the actual data (ESP). Verified with show crypto ipsec sa, where the encaps/decaps counters should increase.
- **Interesting traffic:** the tunnel is triggered only when traffic matches the crypto ACL, which must be a mirror image on both routers.

1.3 NAT/PAT together with a VPN

The edge router performs PAT so private hosts can reach the Internet through one public address. VPN traffic between the branches, however, must not be translated: NAT runs before the crypto ACL is checked, so a translated packet no longer matches the crypto ACL and never enters the tunnel. The fix is a NAT exemption: the NAT ACL first denies branch-to-branch traffic, then permits everything else. This single detail is the cause of many "Internet works but VPN does not" faults.

2. Topology and Addressing

Devices: four Cisco 2911 routers (R1, R2, ISP), two 2960 switches, five PCs, and one server that represents a public website (enable Services > HTTP).

```
Branch A LAN 192.168.10.0/24
  PC-A1, PC-A2, PC-A3 --- SW-A --- Gi0/0 [R1] Gi0/1 ---+
                                                       |
```

```
                                                    [ ISP ] --- Gi0/2 --- Server
                                                      |          198.51.100.10
  PC-B1, PC-B2 --------- SW-B --- Gi0/0 [R2] Gi0/1 ---+
Branch B LAN 192.168.20.0/24

R1 <====== IPsec VPN tunnel (logical, through the ISP) ======> R2
```

Device	Interface	IP Address	Mask	Gateway
R1 (Branch A edge)	Gi0/0 (inside)	192.168.10.1	255.255.255.0	n/a
R1	Gi0/1 (outside)	209.165.200.2	255.255.255.252	209.165.200.1
ISP	Gi0/0	209.165.200.1	255.255.255.252	n/a
ISP	Gi0/1	209.165.201.1	255.255.255.252	n/a
ISP	Gi0/2	198.51.100.1	255.255.255.0	n/a
R2 (Branch B edge)	Gi0/1 (outside)	209.165.201.2	255.255.255.252	209.165.201.1
R2	Gi0/0 (inside)	192.168.20.1	255.255.255.0	n/a
PC-A1 / A2 / A3	NIC	192.168.10.10 / .11 / .12	255.255.255.0	192.168.10.1
PC-B1 / B2	NIC	192.168.20.10 / .11	255.255.255.0	192.168.20.1
Web Server	NIC	198.51.100.10	255.255.255.0	198.51.100.1

Gateway column for router rows shows the default route next hop. The ISP does not route private (192.168.x.x) networks, which is why NAT and the VPN are needed. All addresses are example values.

[Insert screenshot: complete WAN topology with labels]

3. Task 1: Build the WAN, VPN Tunnel and NAT

Step 1: Base configuration

1. Place and cable the devices, then assign the IP addresses from the table. Use straight-through cables to switches and cross-over cables router to router or router to server.
2. ISP router (no NAT, no VPN, just connectivity):

```
Router(config)# hostname ISP
ISP(config)# interface g0/0
ISP(config-if)# ip address 209.165.200.1 255.255.255.252
ISP(config-if)# no shutdown
ISP(config-if)# interface g0/1
ISP(config-if)# ip address 209.165.201.1 255.255.255.252
ISP(config-if)# no shutdown
ISP(config-if)# interface g0/2
ISP(config-if)# ip address 198.51.100.1 255.255.255.0
ISP(config-if)# no shutdown
```

3. Enable the security technology package on R1 and R2 (needed for crypto commands on 2911 routers; the exact steps can differ slightly between Packet Tracer versions):

```
R1(config)# license boot module c2900 technology-package securityk9
R1(config)# end
```

```
R1# copy running-config startup-config
R1# reload
R1# show version        ! confirm securityk9 is Enabled
```

Step 2: Edge router R1 (Branch A)

```
! Interfaces and default route
R1(config)# interface g0/0
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ip nat inside
R1(config-if)# no shutdown
R1(config-if)# interface g0/1
R1(config-if)# ip address 209.165.200.2 255.255.255.252
R1(config-if)# ip nat outside
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# ip route 0.0.0.0 0.0.0.0 209.165.200.1

! NAT with exemption for VPN traffic, then PAT (overload)
R1(config)# ip access-list extended 101
R1(config-ext-nacl)# 10 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
R1(config-ext-nacl)# 20 permit ip 192.168.10.0 0.0.0.255 any
R1(config-ext-nacl)# exit
R1(config)# ip nat inside source list 101 interface g0/1 overload

! IPsec VPN
R1(config)# crypto isakmp policy 10
R1(config-isakmp)# encryption aes 256
R1(config-isakmp)# hash sha
R1(config-isakmp)# authentication pre-share
R1(config-isakmp)# group 2
R1(config-isakmp)# lifetime 86400
R1(config-isakmp)# exit
R1(config)# crypto isakmp key WanKey123 address 209.165.201.2
R1(config)# crypto ipsec transform-set TS esp-aes 256 esp-sha-hmac
R1(cfg-crypto-trans)# exit
R1(config)# access-list 110 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
R1(config)# crypto map VPN-MAP 10 ipsec-isakmp
R1(config-crypto-map)# set peer 209.165.201.2
R1(config-crypto-map)# set transform-set TS
R1(config-crypto-map)# match address 110
R1(config-crypto-map)# exit
R1(config)# interface g0/1
R1(config-if)# crypto map VPN-MAP
```

Step 3: Edge router R2 (Branch B), the mirror image

```
R2(config)# interface g0/0
R2(config-if)# ip address 192.168.20.1 255.255.255.0
R2(config-if)# ip nat inside
R2(config-if)# no shutdown
R2(config-if)# interface g0/1
R2(config-if)# ip address 209.165.201.2 255.255.255.252
R2(config-if)# ip nat outside
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# ip route 0.0.0.0 0.0.0.0 209.165.201.1
R2(config)# ip access-list extended 101
R2(config-ext-nacl)# 10 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
R2(config-ext-nacl)# 20 permit ip 192.168.20.0 0.0.0.255 any
R2(config-ext-nacl)# exit
R2(config)# ip nat inside source list 101 interface g0/1 overload

R2(config)# crypto isakmp policy 10
```

```
R2(config-isakmp)# encryption aes 256
R2(config-isakmp)# hash sha
R2(config-isakmp)# authentication pre-share
R2(config-isakmp)# group 2
R2(config-isakmp)# lifetime 86400
R2(config-isakmp)# exit
R2(config)# crypto isakmp key WanKey123 address 209.165.200.2
R2(config)# crypto ipsec transform-set TS esp-aes 256 esp-sha-hmac
R2(cfg-crypto-trans)# exit
R2(config)# access-list 110 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
R2(config)# crypto map VPN-MAP 10 ipsec-isakmp
R2(config-crypto-map)# set peer 209.165.200.2
R2(config-crypto-map)# set transform-set TS
R2(config-crypto-map)# match address 110
R2(config-crypto-map)# exit
R2(config)# interface g0/1
R2(config-if)# crypto map VPN-MAP
```

Step 4: Verify

1. From PC-A1 ping PC-B1 (192.168.20.10). The first ping or two may time out while the tunnel is negotiated; later pings should succeed.
2. From PC-A1 and PC-B1, open a browser and visit http://198.51.100.10 (both branches reach the Internet).
3. On R1, check the tunnel:

```
R1# show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status
209.165.201.2   209.165.200.2   QM_IDLE           1001 ACTIVE

R1# show crypto ipsec sa
   #pkts encaps: 8, #pkts encrypt: 8, #pkts decaps: 8, #pkts decrypt: 8
```

Outputs in this document are typical examples. Counters, IDs and some wording will differ in your Packet Tracer version.

[Insert screenshot: successful ping between branches and show crypto isakmp sa on R1]

4. Task 2: PAT for Three Devices Using One Public IP

Branch A has three internal devices (PC-A1, PC-A2, PC-A3). The overload keyword in the NAT statement from Task 1 already lets all of them share the single public address of R1's outside interface (209.165.200.2):

```
R1(config)# ip nat inside source list 101 interface g0/1 overload
```

1. Clear old entries: R1# clear ip nat translation *
2. Open a browser on PC-A1, PC-A2 and PC-A3 and visit http://198.51.100.10.
3. Capture the table on R1:

```
R1# show ip nat translations
Pro  Inside global         Inside local        Outside local       Outside global
tcp  209.165.200.2:1025    192.168.10.10:1025  198.51.100.10:80    198.51.100.10:80
tcp  209.165.200.2:1026    192.168.10.11:1025  198.51.100.10:80    198.51.100.10:80
tcp  209.165.200.2:1027    192.168.10.12:1025  198.51.100.10:80    198.51.100.10:80
```

Observation: three private addresses map to one public address (209.165.200.2) with different port numbers. Without the overload keyword, only one host could use the address at a time and the others would fail. The deny line in ACL 101 keeps branch-to-branch traffic out of this table.

[Insert screenshot: browser on each PC showing the web page, and the NAT table]

5. Task 3: Misconfigured Pre-Shared Key

5.1 Introduce the fault

On R2, replace the correct key with a wrong one (R1 keeps WanKey123):

```
R2(config)# no crypto isakmp key WanKey123 address 209.165.200.2
R2(config)# crypto isakmp key WrongKey99 address 209.165.200.2
R2# clear crypto isakmp
R2# clear crypto sa
```

5.2 Symptom

PC-A1 can still browse http://198.51.100.10, but ping 192.168.20.10 fails with Request timed out. The fault is therefore in the VPN, not in basic routing or NAT.

5.3 Diagnose with show and debug

1. Check Phase 1 on R1 after generating traffic with a ping:

```
R1# show crypto isakmp sa
dst             src             state          conn-id status
209.165.201.2   209.165.200.2   MM_NO_STATE       0    ACTIVE
```

Phase 1 never reaches QM_IDLE, so authentication or the IKE policy is failing. Phase 2 will not start. (If the table is empty, check that interesting traffic is actually being generated.)

2. Watch the negotiation in real time:

```
R1# debug crypto isakmp
  ... send a ping from PC-A1 to PC-B1 ...
%CRYPTO-4-IKMP_BAD_MESSAGE: IKE message from 209.165.201.2 failed its
sanity check or is malformed
R1# undebug all
```

A bad-message or sanity-check error on Phase 1 after the peers exchange their first packets is the typical signature of a pre-shared key mismatch. Because the key is used to decrypt and verify the peer's message, the wrong key makes it look corrupted.

3. Compare the configuration on both routers. The key itself is shown in clear text on these routers, so a direct comparison is possible:

```
R1# show running-config | include crypto isakmp key
crypto isakmp key WanKey123 address 209.165.201.2

R2# show running-config | include crypto isakmp key
crypto isakmp key WrongKey99 address 209.165.200.2
```

The keys differ, so this is the root cause. (Also confirm the peer address on each line is the other router's outside IP and that both isakmp policies match.)

5.4 Fix and verify

```
R2(config)# no crypto isakmp key WrongKey99 address 209.165.200.2
```

```
R2(config)# crypto isakmp key WanKey123 address 209.165.200.2
R2(config)# end
R2# clear crypto isakmp
R2# clear crypto sa

R1# show crypto isakmp sa          ! after a new ping: state QM_IDLE, ACTIVE
R1# show crypto ipsec sa           ! encaps/decaps counters now increase
```

Result: ping from PC-A1 to PC-B1 succeeds again, and Phase 1 shows QM_IDLE.

[Insert screenshot: MM_NO_STATE before the fix and QM_IDLE after the fix]

6. Task 4: Internet Works but the Other Branch Is Unreachable

6.1 Introduce the fault

Remove the NAT exemption on R1 so that VPN traffic is translated before it can be matched by the crypto ACL:

```
R1(config)# ip access-list extended 101
R1(config-ext-nacl)# no 10
R1# clear ip nat translation *
R1# clear crypto sa
```

6.2 Symptom

- PC-A1 ping 198.51.100.10 succeeds and the web page loads, so the Internet path, routing and PAT are fine.
- PC-A1 ping 192.168.20.10 fails.

6.3 Troubleshooting commands and findings

Command	What it shows	Interpretation
show crypto isakmp sa (R1)	No SA, or no QM_IDLE entry	Phase 1 not established because the tunnel was never triggered
show crypto ipsec sa (R1)	#pkts encaps: 0	No packets are matching the crypto ACL
show access-lists 110 (R1)	permit ip 192.168.10.0 ... (0 matches)	Crypto ACL never matched, so traffic is being changed before it reaches the crypto map
show ip nat translations (R1)	icmp 209.165.200.2:5 192.168.10.10:5 192.168.20.10:5 ...	Packets for 192.168.20.x are being NATed. This is the key clue.
show access-lists 101 (R1)	Only the permit line remains	The deny (exemption) line is missing
debug ip nat (R1)	NAT*: s=192.168.10.10->209.165.200.2, d=192.168.20.10	Confirms translation is happening to VPN-bound traffic

Root cause: the NAT ACL permits all inside traffic, including traffic for the other branch. The source address is rewritten to the public IP before the crypto ACL check, so the packet no longer matches ACL 110 and is sent unencrypted toward the Internet, where private addresses are dropped by the ISP.

6.4 Configuration change that resolves it

```
R1(config)# ip access-list extended 101
R1(config-ext-nacl)# 10 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
R1(config-ext-nacl)# end
R1# clear ip nat translation *

R1# show access-lists 101           ! deny line before permit line
R1# show crypto isakmp sa           ! after a new ping: QM_IDLE
R1# show crypto ipsec sa            ! encaps/decaps increasing
```

Result: PC-A1 can ping PC-B1 and still browse the Internet, since the permit line continues to apply PAT to everything else.

6.5 Other causes to rule out in the same symptom

Possible cause	How to detect it
Crypto map not applied to the outside interface	show run | section interface: no crypto map line under g0/1; show crypto map
Crypto ACLs not mirror images (wrong subnet or mask)	show access-lists 110 on both routers; debug crypto isakmp shows proxy identities not supported in Phase 2
Transform-set or ISAKMP policy mismatch	show crypto isakmp policy and show crypto ipsec transform-set on both routers
Wrong peer address in crypto map	show crypto map; compare set peer with the other router's outside IP
Missing default route to the ISP	show ip route: no 0.0.0.0/0; ping the peer's outside IP from the router
Missing NAT exemption on the remote router	Same checks as above on R2; traffic leaving B must also skip NAT

[Insert screenshot: show access-lists 101, show ip nat translations and the fixed tunnel]

7. Task 5: Peer Review of a Classmate's Configuration

7.1 Procedure

1. Save your Packet Tracer file (.pkt) and exchange it with a classmate. If you cannot share the file, copy the output of show running-config from each router into a text file and swap that.
2. Open their file and test: PC to PC across branches, PC to the web server, show crypto isakmp sa, show crypto ipsec sa, show ip nat translations.
3. Use show and debug commands to locate the fault. Do not guess or start editing before you have evidence.
4. Fix one misconfiguration, retest, and write a short note using the template below.

7.2 Review checklist (work in this order)

- Layer 1/2: interfaces up/up (show ip interface brief).
- Layer 3: IP addresses, default routes, ping the peer's outside address.
- NAT: ip nat inside/outside on the right interfaces; NAT ACL denies branch-to-branch traffic first.

- VPN Phase 1: isakmp policy, key and peer address match on both routers.
- VPN Phase 2: transform sets match; crypto ACLs are mirror images; crypto map applied to the outside interface.

7.3 Review note template with a worked example

The example below shows how to fill the note. Replace it with the real issue you find in your classmate's setup.

Field	Entry (example)
Reviewer / Owner	[your name] / [classmate's name]
Symptom	Hosts in Branch B reach the web server but cannot ping Branch A.
Commands used	show crypto isakmp sa (QM_IDLE, up); show crypto ipsec sa (encaps rising on R2, decaps 0); show access-lists 110 on both routers; debug crypto isakmp
Evidence	Phase 1 was up, so the key was fine. R1 ACL 110 permitted 192.168.10.0/24 to 192.168.20.0/24, but R2 ACL 110 listed 192.168.200.0/24 as the destination (typo). debug crypto isakmp showed Phase 2 failing because the proxy identities did not match.
Root cause	Crypto ACLs on R1 and R2 were not mirror images.
Fix	On R2: no access-list 110, then access-list 110 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255; clear crypto sa
Verification	Ping across the tunnel works; show crypto ipsec sa shows encaps and decaps increasing on both routers.

[Insert screenshot: your own peer-review evidence (show/debug output of the fault and the fix)]

8. Checkpoint Quiz: WAN, VPN and NAT/PAT

1. Which WAN option is a dedicated point-to-point circuit with guaranteed bandwidth?

A. Broadband cable
B. Leased line
C. Wi-Fi
D. Public hotspot

2. Which IPsec protocol provides confidentiality (encryption) of the data?

A. ESP
B. ARP
C. ICMP
D. DHCP

3. What is the purpose of IKE Phase 1?

A. Encrypt user data
B. Authenticate peers and build a secure management channel
C. Assign IP addresses
D. Translate ports

4. Which command displays the Phase 1 security association?

A. show ip route
B. show crypto isakmp sa
C. show vlan brief
D. show ip nat statistics

5. Which state indicates a healthy Phase 1 SA?

A. MM_NO_STATE
B. QM_IDLE
C. DOWN
D. INIT

6. Why is a NAT exemption needed on VPN edge routers?

A. To save memory
B. So branch-to-branch traffic is not translated before the crypto ACL is checked
C. To speed up DNS
D. To disable the firewall

7. What does the overload keyword do in a NAT statement?

A. Adds more IP addresses
B. Lets many inside hosts share one public IP using port numbers
C. Blocks outside traffic
D. Enables IPv6

8. In PAT with an interface-based statement, the inside global address is:

A. The PC's private IP
B. The router's outside interface IP
C. The ISP's loopback
D. The default gateway of the PC

9. Crypto ACLs on the two VPN peers must be:

A. Identical
B. Mirror images of each other
C. Empty
D. Applied to the LAN interface only

10. A pre-shared key mismatch typically leaves the Phase 1 SA in which state?

A. QM_IDLE
B. MM_NO_STATE
C. ESTABLISHED
D. FULL

Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	A	B	B	B	B	B	B	B	B

9. Practical Evaluation Checklist

#	Item to demonstrate	Evidence / command	Done
1	Addressing and ISP connectivity	show ip interface brief; ping ISP from R1 and R2	[ ]
2	PAT works for three devices	show ip nat translations with three inside locals	[ ]

#	Item to demonstrate	Evidence / command	Done
3	NAT exemption configured	show access-lists 101: deny before permit	[ ]
4	VPN tunnel up	show crypto isakmp sa = QM_IDLE; ipsec counters rise	[ ]
5	Branch-to-branch ping	PC-A1 to PC-B1 reply	[ ]
6	Internet access from both branches	Browser to 198.51.100.10 from A and B	[ ]
7	Fault found with show/debug (PSK)	MM_NO_STATE, debug output, fixed key	[ ]
8	Fault found with show/debug (NAT exemption)	ipsec encaps 0, NAT table clue, fixed ACL	[ ]
9	Configuration saved	copy running-config startup-config on all routers	[ ]

10. Troubleshooting Cheat Sheet and Conclusion

Symptom	Command to run	Likely cause
No ping to peer outside IP	show ip route; show ip interface brief	Missing default route, interface down, wrong IP
Internet works, VPN does not	show ip nat translations; show access-lists	Missing NAT exemption or no crypto map on interface
Phase 1 MM_NO_STATE	debug crypto isakmp; show run | include isakmp	Key, policy or peer mismatch
Phase 1 up, no data	show crypto ipsec sa; show access-lists 110	Crypto ACL mismatch, transform-set mismatch
Some hosts work, others do not	show access-lists 101; show ip nat statistics	NAT ACL missing a subnet, PAT exhaustion

Conclusion: in this exercise I built a WAN with two branch offices joined by an IPsec site-to-site VPN, configured PAT on both edge routers, and then diagnosed two realistic faults, a mismatched pre-shared key and a missing NAT exemption. In both cases the answer came from reading show and debug output in order (interfaces, routing, NAT, Phase 1, Phase 2) rather than from guessing. This method works the same on real routers.